Google Play Store App Insights: Solution Guide

Prepared by: S. Geetanjali **Tool:** Microsoft Power BI Desktop (Power Query and DAX)
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1. Executive Summary

This report analyses 9,658 unique apps from the Google Play Store, together with 37,432 labelled user reviews, to understand what drives app success and how a mobile app company can use those findings.

Key findings

- The Play Store is dominated by free apps (92.2%), and free apps attract about 27 times more reviews per app than paid apps.
- The average rating is 4.17, and 76.7% of rated apps score 4.0 or above.
- Popularity does not predict quality: the correlation between installs and rating is only 0.04.
- Game, Communication, Tools, Family and Productivity lead on installs, while Events, Art & Design, Education and Books & Reference lead on ratings.
- Apps priced above $10 have the lowest ratings among paid apps (4.13), while paid apps under $5 rate 4.26 to 4.30.
- Highly rated apps receive 65.5% positive reviews against 54.0% for lower-rated apps, and their average sentiment polarity is twice as high (0.192 vs 0.096).

2. Problem Statement and Objectives

A mobile application company has acquired a Play Store dataset and wants evidence to improve its app development strategy, user experience, installs and revenue. The analysis answers three questions:

1. What key factors contribute to an app's success on the Google Play Store?
2. How can the company leverage these factors to improve its app offerings?
3. What can be learned about user preferences and behaviour?

Stakeholders. Internal: app developers, product managers, the marketing team and senior management. External: app users, advertisers and partners.

3. Data and Data Dictionary

Sources: `googleplaystore.xlsx` (10,841 rows, 13 columns) and `googleplaystore_user_reviews.xlsx` (64,295 rows, 5 columns).

Column	Description
App	Name of the app
Category	Category of the app (33 categories)
Rating	Average user rating, 0 to 5
Reviews	Number of user reviews
Size	Size of the app (converted to Size_MB)
Installs	Install bucket, such as 1,000+ (converted to a number)
Type	Free or Paid
Price	Price in US dollars
Content Rating	Age group the app is suitable for
Genres	Genre or genres of the app
Last Updated	Date of the most recent update
Current Ver	Current app version
Android Ver	Minimum Android version required

Reviews file columns: App, Translated_Review, Sentiment (Positive, Negative, Neutral), Sentiment_Polarity (-1 to +1) and Sentiment_Subjectivity.

4. Data Cleaning and Preparation

All cleaning was done in Power Query so that every step is documented and repeatable.

Step	Action	Rows after
1	Loaded the raw file	10,841
2	Removed 1 corrupted row (Category = 1.9, values shifted left)	10,840
3	Removed 483 exact duplicate rows	10,357

Step	Action	Rows after
4	Removed 1 row with an invalid Type value	10,356
5	Sorted by Reviews and kept one row per app name (the most-reviewed entry)	9,658

Type conversions and standardisation

- **Installs:** removed `+` and `,`, then converted to a whole number.
- **Price:** removed `$`, then converted to a decimal number.
- **Rating:** replaced the text NaN with null (about 15% of apps are unrated) and converted to decimal. Nulls were kept, not set to 0, so averages are not distorted.
- **Size:** created Size_MB. Sizes in k were divided by 1,024, and 'Varies with device' was set to null (1,227 apps).
- **Reviews:** converted to a whole number. **Last Updated:** converted to a date.
- **Text:** underscores removed and Category capitalised (for example, ART_AND_DESIGN became Art And Design). Missing Current Ver values were set to 'Unknown'.
- **Reviews file:** 26,863 rows without a sentiment label were removed, leaving 37,432 labelled reviews covering 865 apps. The reviews were merged with the apps table on exact app name, which matched 33,941 review rows (90.7%).

5. Success Metrics

Metric	Definition
Total Apps	Number of unique apps
Avg Rating	Mean rating across rated apps
Total Installs / Avg Installs	Sum and mean of the install bucket value
Avg Reviews	Mean number of reviews per app, a measure of engagement
Apps Rated 4+	Apps with a rating of 4.0 or higher
Review Count and Avg Polarity	Number of labelled reviews and mean sentiment polarity

6. Basic Questions (Q1 to Q10)

Q	Question	Answer	Insight and business impact
Q1	Average rating	**4.17** (8,196 rated apps)	Sets the quality benchmark; a new app should aim above 4.2.
Q2	Unique categories	**33**	A wide market with room for niche opportunities.
Q3	Size distribution	Median **12 MB**, mean **20.4 MB**; most apps are under 20 MB	Right-skewed: a few large games pull the mean up. Keep apps lean unless the category demands size.
Q4	Free vs paid	**8,904 free (92.2%)** and **754 paid (7.8%)**	Free is the norm; paid apps must justify their price.
Q5	Most common content rating	**Everyone**, 7,903 apps (about 82%), then Teen (about 11%)	Designing for all ages reaches the largest audience.
Q6	Top 5 most installed	Facebook, WhatsApp Messenger, Instagram, Messenger and Subway Surfers (all 1B+ installs)	20 apps tie at 1B+ installs, so reviews were used as the tiebreaker.
Q7	Apps rated 4.0+	**6,287** (76.7% of rated apps)	Most apps are well rated, so a 4.0 rating is only par.
Q8	Average reviews, free vs paid	Free about **234,447**, paid about **8,748**	Free apps engage far more users, and paid apps need active review prompts.
Q9	Average size by category	Largest: Game (**41.7 MB**), Family (27 MB). Smallest: Tools (**8.8 MB**)	Plan storage and performance budgets by category.
Q10	Apps updated in 2018	**6,286** (65% of apps)	Regular updates are the norm in this market.

[Insert screenshot: Basic Insights page]

7. Medium Questions (M1 to M10)

Q	Question	Answer	Insight and business impact
M1	Installs vs rating correlation	**0.04**	Essentially no relationship, so chasing installs will not improve ratings.
M2	Highest-rated categories	Events **4.44**, Art And Design **4.36**, Education **4.36**, Books And Reference **4.34**, Personalization **4.33**. Lowest: Dating **3.98**	Niche categories delight users; Dating is the weakest.
M3	Price vs rating (paid apps)	$0.01-1: 4.30, $1-2: 4.29, $2-5: 4.26, $5-10: 4.26, $10+: **4.13**	Higher prices raise expectations; apps over $10 rate lowest.
M4	Rating by content rating	Adults only 18+ 4.30, Everyone 10+ 4.23, Teen 4.23, Everyone 4.17, Mature 17+ 4.12, Unrated 4.10	Differences are small. Adults only and Unrated have very few apps.
M5	Genres with the most 1M+ install apps	Tools **274**, Action **180**, Photography **173**, Entertainment **154**, Productivity **147**	These genres have proven demand but strong competition.
M6	Update frequency	Average **281 days** since last update (relative to 8 Aug 2018)	Only the latest update is stored, so true update intervals cannot be measured.
M7	Size vs installs	Installs rise with size: apps under 5 MB average about 0.76M installs, apps over 50 MB about 13M	Larger apps are usually richer, popular products (mostly games).
M8	Most-reviewed apps and ratings	Facebook (78.2M reviews, 4.1), WhatsApp (69.1M, 4.4), Instagram (66.6M, 4.5), Messenger (56.6M, 4.0), Clash of Clans (44.9M, 4.6)	Huge engagement does not require a perfect rating.

Q	Question	Answer	Insight and business impact
M9	Content rating mix, free vs paid	Everyone: about 81% of free and 87% of paid apps. Teen: about 11% of free and 7% of paid	Free apps skew slightly older, paid apps skew family-friendly.
M10	Top 5 categories by installs	Game **13.5B**, Communication **11.0B**, Tools **8.1B**, Family **6.2B**, Productivity **5.8B**	Where demand is greatest.

[Insert screenshot: Medium Insights page]

8. Advanced Questions (A1 to A5)

A1. Top 10 highest-rated apps

271 apps have a perfect 5.0 rating, but their median is only 4 reviews, so the ranking is not meaningful. Applying a minimum of 1,000 reviews and a rating of at least 4.9 leaves 23 apps. The top 10 by reviews are:

App	Rating	Reviews	Installs
JW Library	4.9	922,752	10,000,000+
Six Pack in 30 Days - Abs Workout	4.9	272,337	10,000,000+
Tickets + PDA 2018 Exam	4.9	197,136	1,000,000+
Learn Japanese, Korean, Chinese Offline & Free	4.9	133,136	1,000,000+
StrongLifts 5x5 Workout Gym Log & Personal Trainer	4.9	66,791	1,000,000+
PixPanda - Color by Number Pixel Art Coloring Book	4.9	55,723	1,000,000+
ipsy: Makeup, Beauty, and Tips	4.9	49,790	1,000,000+
Hungry Hearts Diner: A Tale of Star-Crossed Souls	4.9	46,253	500,000+
Lose Belly Fat in 30 Days - Flat Stomach	4.9	38,098	5,000,000+
Solitaire: Decked Out Ad Free	4.9	37,302	500,000+

Insight: the best-rated apps are focused utilities, fitness, learning and hobby apps with strong engagement and at least 500,000 installs, so high quality and large reach can go together.

A2. Trend of app updates over time

Year	2010	2011	2012	2013	2014	2015	2016	2017	2018
Apps last updated	1	15	26	108	203	448	779	1,792	6,286

In 2018 the monthly counts rise from 286 in January to a peak of **2,318 in July**, then 983 in August (a partial month). Across all years, July is the busiest month (2,610 apps).

Caveat: each app stores only its latest update, and the data ends on 8 August 2018. The recent spike therefore reflects the data snapshot as much as real seasonality, and true update frequency would need historical update logs.

A3. Average rating by install band

Install band	<1K	1K-10K	10K-100K	100K-1M	1M-10M	10M+
Avg rating	4.35	4.05	4.04	4.13	4.22	4.32

Ratings follow a U shape. Very small apps have few raters, which can inflate scores, while mid-sized apps are rated lowest and the most-installed apps recover to 4.32. This reinforces the M1 finding that installs and ratings are only weakly linked.

A4. Sentiment analysis

Of 37,432 labelled reviews, 23,998 (64.1%) are positive, 8,271 (22.1%) negative and 5,163 (13.8%) neutral.

Rating band	Positive	Negative	Neutral	Avg polarity
High (4.0+)	65.5%	21.6%	12.9%	**0.192**
Low (below 4.0)	54.0%	27.6%	18.4%	**0.096**

Negative reviews are concentrated in popular games: Angry Birds Classic (147), Candy Crush Saga (126), Bowmasters (119), 8 Ball Pool (106) and Candy Crush Soda Saga (96). The most positive reviews go to Helix Jump (209), Duolingo (200) and Calorie Counter - Macros (174). The sentiment labels confirm that star ratings reflect real user feedback. Only 865 apps have reviews, so this analysis covers a subset of the market.

A5. Genre and user ratings

The dataset has 118 distinct genres (multi-genre labels count separately). Considering genres with at least 30 rated apps:

Genre	Avg rating	Median rating
Events	4.44	4.5
Puzzle	4.37	4.4
Art & Design	4.36	4.4
Books & Reference	4.34	4.5
Parenting	4.34	4.5
Maps & Navigation (low)	4.04	4.2
Dating (low)	3.98	4.1
Educational (lowest)	3.87	3.95

Puzzle, Events and Art & Design are consistently high; Dating and Maps & Navigation are consistently low. In the source labels, 'Education' and 'Education;Education' appear as separate genres.

[Insert screenshot: Advanced Insights pages and Executive Summary]

9. Key Insights and Recommendations

Insights

1. The market is overwhelmingly free, with large audiences for free apps and very limited engagement for paid ones.
2. Installs and ratings are almost unrelated, so quality has to be earned separately from popularity.
3. Demand is concentrated in Game, Communication, Tools, Family and Productivity, while ratings peak in niches such as Events, Art & Design, Education and Books.
4. Price matters: paid apps do slightly better than free on rating, except above $10.
5. Satisfied users write warmer reviews, and games draw the most negative feedback in absolute numbers.

Recommendations by stakeholder

Stakeholder	Recommendation	Evidence
Senior management	Use a free model (ads or in-app purchases) for reach, and keep any paid app under $5.	92.2% of apps are free; ratings fall above $10.
Product managers	Pair a high-install category (for volume) with a high-rated niche (for quality and lower competition).	M2, M5, M10
App developers	Keep apps within category size norms (Tools about 9 MB, Games about 42 MB) and update regularly.	Q9, Q10
App developers	Monitor review sentiment and fix recurring complaints to move from the low rating band to the high band.	A4
Marketing team	Target the Everyone content rating for the widest audience, and encourage reviews on paid apps.	Q5, Q8
Partners and advertisers	Focus on high-install categories (Game, Communication, Tools) for reach.	M10

10. Limitations and Ethical Considerations

- **Missing values:** about 15% of apps have no rating and 1,227 apps have no size; these were left as blanks rather than filled.
- **Install buckets:** Installs are lower-bound categories (for example 1,000,000+), so totals are approximate.
- **One row per app:** the most-reviewed entry was kept for each app. About 21 app names differ only by capitalisation and were treated as separate apps.
- **Snapshot data:** the data ends on 8 August 2018 and stores only the latest update, so update frequency and seasonality cannot be measured directly.
- **Reviews coverage:** only 865 apps have labelled reviews, and 90.7% of review rows matched an app by exact name.
- **Correlation is not causation:** relationships such as size and installs do not prove cause.
- **Privacy:** the data contains public app information and review text, with no personal identifiers; any further use of review text should respect user privacy.

- **Template note:** the 'Additional Considerations' in the case study mention patient and loan data, which do not apply to this dataset.

11. Appendix: Power BI Build Notes

Measures created: Total Apps, Avg Rating, Unique Categories, Free Apps %, Apps Rated 4+, Apps Updated 2018, Avg Reviews, Avg Size MB, Avg Installs, Total Installs, Apps 1M+ Installs, Avg Days Since Update, Correlation Installs Rating, Median Rating, Review Count and Avg Polarity.

Calculated columns: Price Band, Install Band, Rating Band (all with sort-order helper columns where needed).

Report pages: Basic Insights, Medium Insights, Advanced Insights, Sentiment Analysis (A4) and Executive Summary.

Basic Insights

Q1

Avg Rating

4.2

Q2

Unique categories

33

Q7

apps rated 4+

6.287K

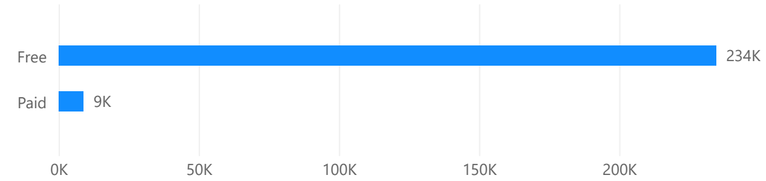
Q10

apps updated 2018

6.286K

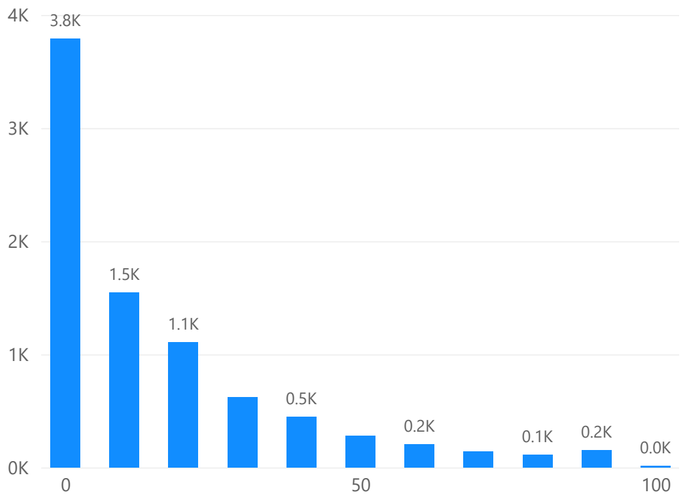
Q8 Avg Review: Free vs. Paid

avg review by Type



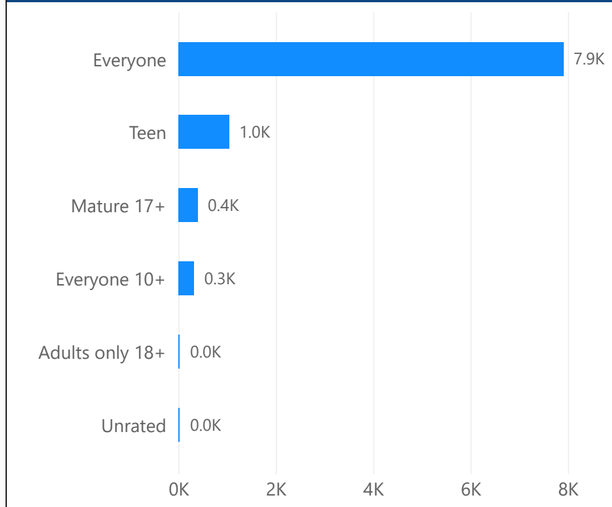
Q9 Average App Size by Category (MB)

Total Apps by Size_MB (bins)



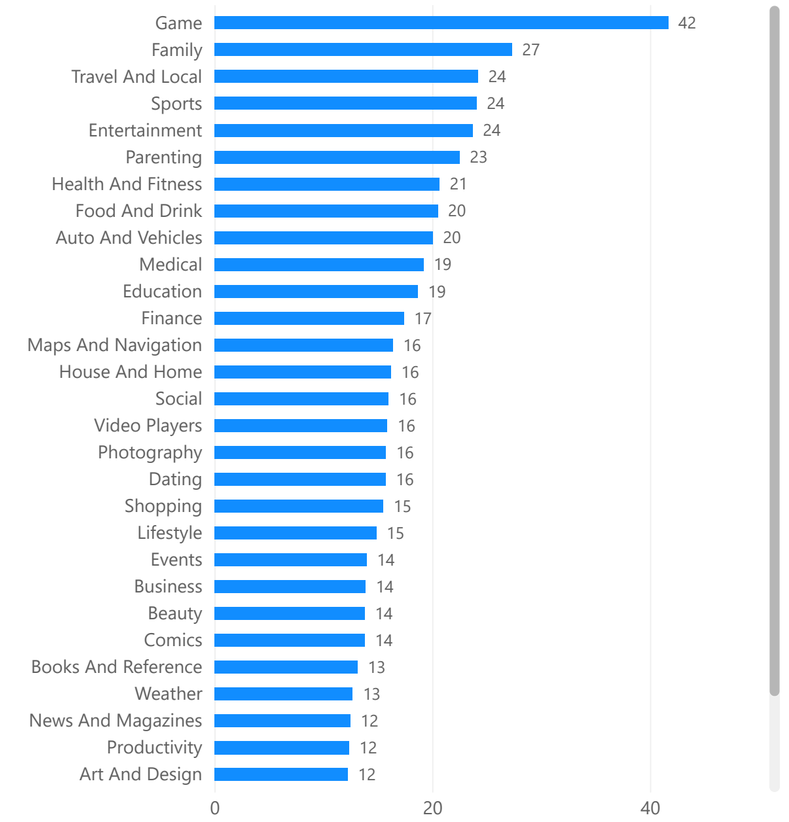
Q5 Apps by content

Total Apps by Content Rating



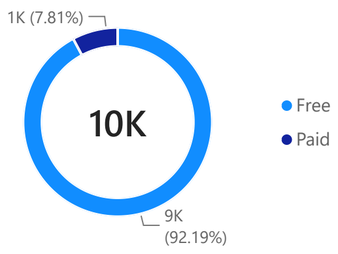
Q3 Avg size MB

avg size MB by Category



Q4 Free vs. Paid app

Total Apps by Type



Q6 Top 5 Most Installed Apps

App	Sum of Installs	Sum of Reviews
Facebook	1000000000	78158306
WhatsApp Messenger	1000000000	69119316
Instagram	1000000000	66577446
Messenger – Text and Video Chat for Free	1000000000	56646578
Subway Surfers	1000000000	27725352

Medium Insights

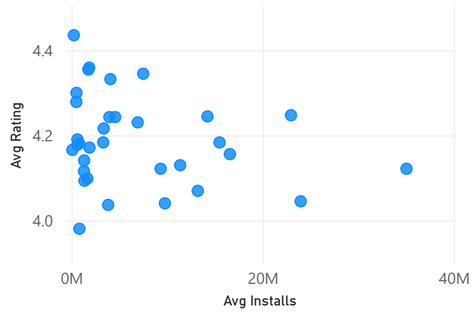
M1

Correlation Installs Rating

0.04

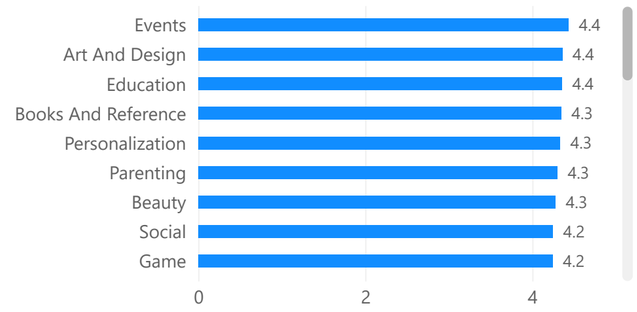
M1 Avg Installs and Avg Rating

Avg Installs and Avg Rating by Category



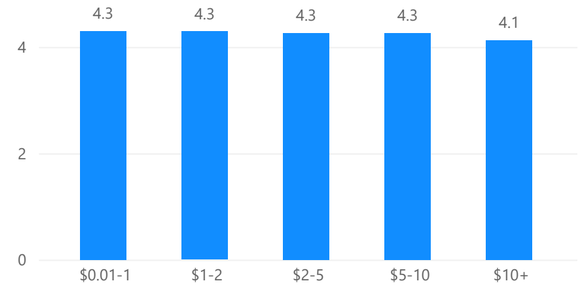
M2 Highest Rated Categories

Avg Rating by Category



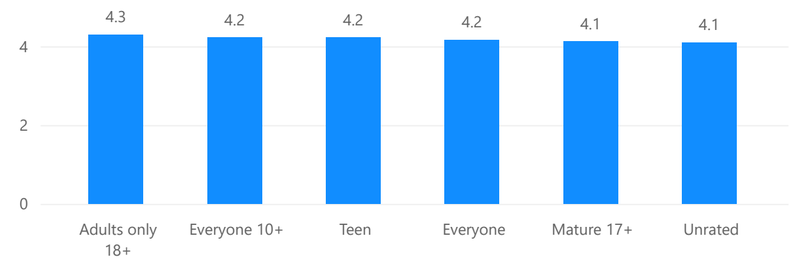
M3 Price vs. Rating

Avg Rating by Price Band



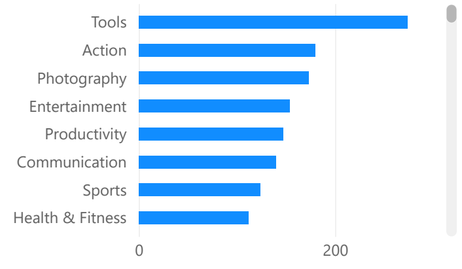
M4 Rating by Content Rating

Avg Rating by Content Rating



M5 Genres with 1M+ Installs

Apps 1M+ Installs by Genres



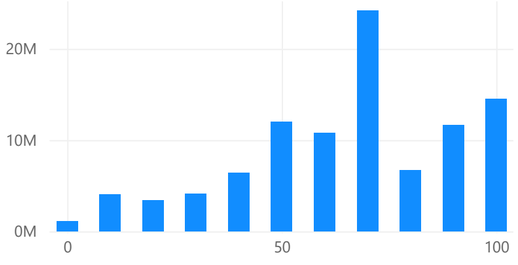
M6 Update Frequency

Avg Days Since Update

281.04

M7 Size vs. Installs

Avg Installs by Size_MB (bins)

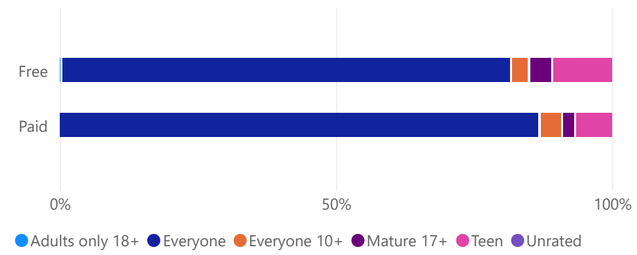


M8 Most Reviewed Apps

App	Sum of Reviews	Sum of Rating
Facebook	78158306	4.10
WhatsApp Messenger	69119316	4.40
Instagram	66577446	4.50
Messenger – Text and Video Chat for Free	56646578	4.00
Clash of Clans	44893888	4.60

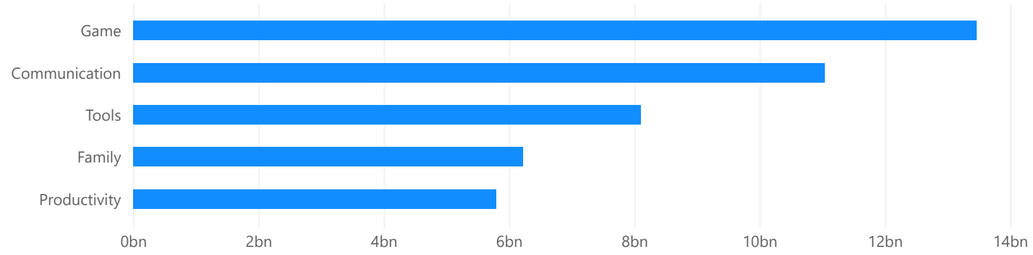
M9 Content rating ,Free vs. Paid

Total Apps by Type and Content Rating



M10 Top 5 Categories By Installs

Total Installs by Category



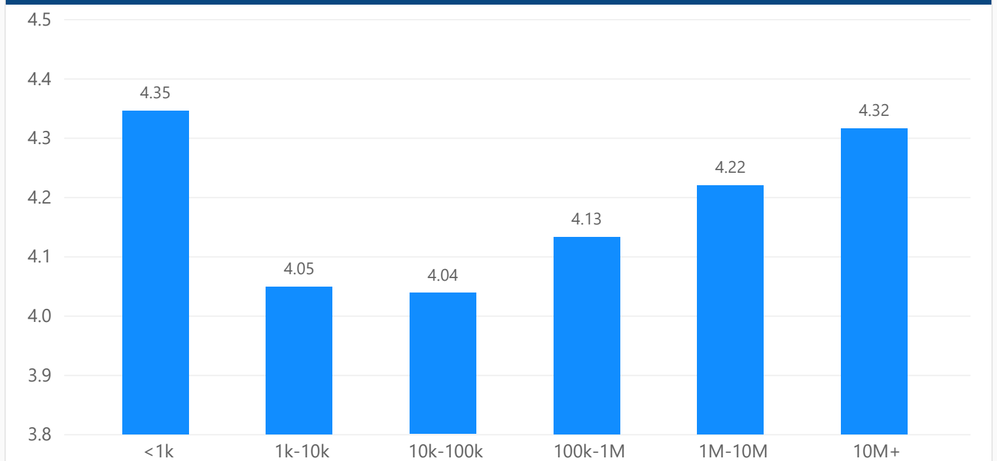
Advanced Insights

A1 Top 10 Rated Apps

App	Average of Rating	Sum of Reviews	Sum of Installs
JW Library	4.90	922752	10000000
Six Pack in 30 Days - Abs Workout	4.90	272337	10000000
Tickets + PDA 2018 Exam	4.90	197136	1000000
Learn Japanese, Korean, Chinese Offline & Free	4.90	133136	1000000
StrongLifts 5x5 Workout Gym Log & Personal Trainer	4.90	66791	1000000
PixPanda - Color by Number Pixel Art Coloring Book	4.90	55723	1000000
ipsy: Makeup, Beauty, and Tips	4.90	49790	1000000
Hungry Hearts Diner: A Tale of Star-Crossed Souls	4.90	46253	500000
Lose Belly Fat in 30 Days - Flat Stomach	4.90	38098	5000000
Solitaire: Decked Out Ad Free	4.90	37302	500000

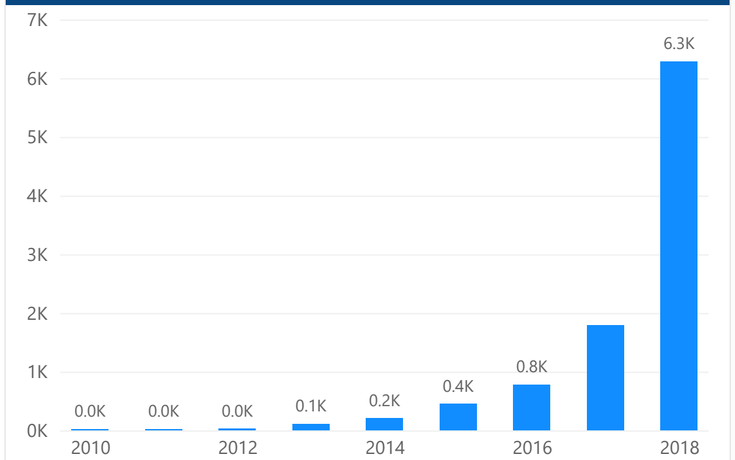
A3 Rating by Installs

Avg Rating by Install Band



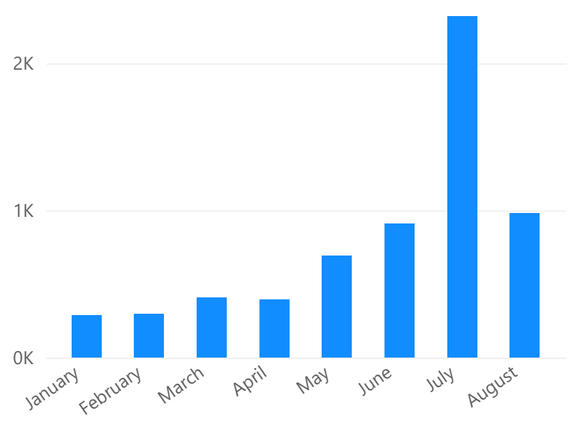
A2 Apps By Last Updated Year

Total Apps by Year



A2 2018 Updates month

Total Apps by Month



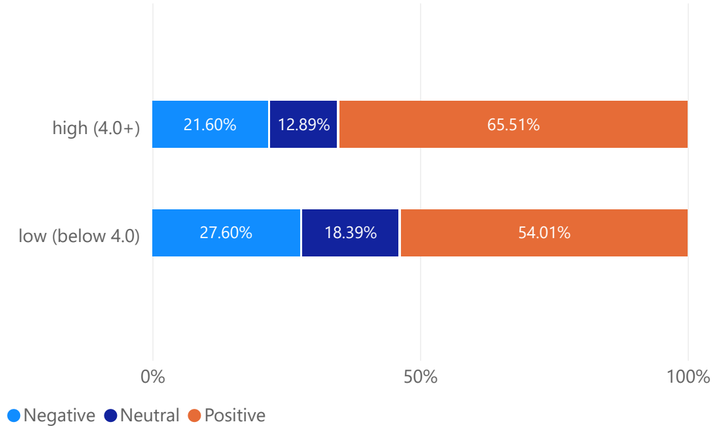
A5 Genre Ratings

Genres	Total Apps	Avg Rating	Median Rating
Events	64	4.44	4.50
Puzzle	119	4.37	4.40
Art & Design	57	4.36	4.40
Books & Reference	222	4.34	4.50
Parenting	46	4.34	4.50
Personalization	376	4.33	4.40
Education;Education	36	4.29	4.40
Education	510	4.29	4.40
Casino	39	4.29	4.40
Board	42	4.28	4.30
Total	**9298**	**4.17**	**4.30**

Advanced Insights

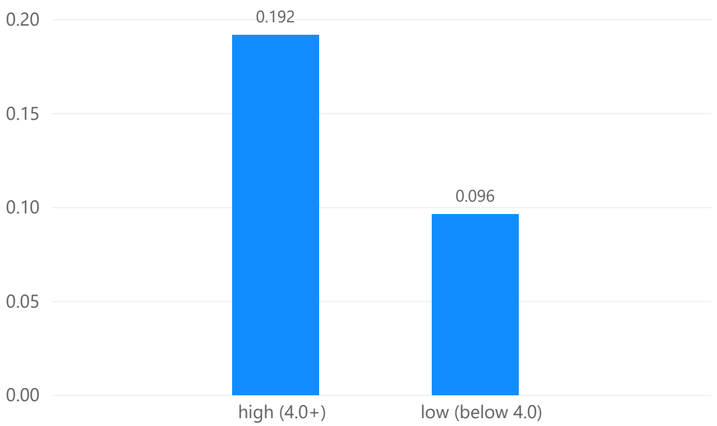
A4 Sentiment Mix By App Rating Band

Review count and Sentiment % by review Rating Band and Sentiment



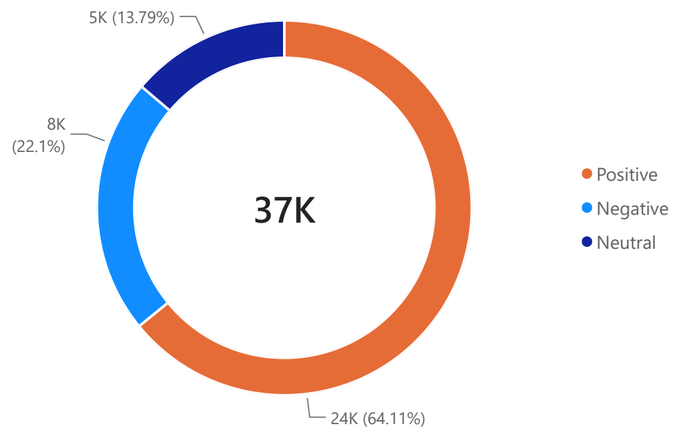
A4 Average Sentiment Polarity By Rating Band

average polarity by review Rating Band



A4 Review count

Review count by Sentiment



Review count

37.432K

A4 Most negative Reviews

App	Review count
Angry Birds Classic	147
Candy Crush Saga	126
Bowmasters	119
8 Ball Pool	106
Candy Crush Soda Saga	96
Garena Free Fire	81
Cooking Fever	79
Alto's Adventure	72
Block Puzzle	71
Agar.io	66
Gardenscapes	63
Facebook	60
Be A Legend: Soccer	59
Clash Royale	58
Helix Jump	58
Expedia Hotels, Flights & Car Rental Travel Deals	55
Calorie Counter - MyFitnessPal	52
Flow Free	49
Episode - Choose Your Story	48
Gyft - Mobile Gift Card Wallet	48
DEAD TARGET: FPS Zombie Apocalypse Survival Games	47
Google Photos	47
Colorfy: Coloring Book for Adults - Free	45
Candy Crush Jelly Saga	44
Farm Heroes Saga	43
Galaxy Attack: Alien Shooter	43
CBS Sports App - Scores, News, Stats & Watch Live	42
Fishdom	42
Fox News – Breaking News, Live Video & News Alerts	42
Basketball Stars	41
Total	**8271**

Key Insights

- 92% of apps are free, and free apps get about 27 times more reviews than paid ones.
- Installs don't predict ratings (correlation 0.04), so popularity isn't quality.
- Game, Communication, Tools, Family and Productivity lead on installs, and Events, Art & Design, Education and Books lead on ratings.
- Paid apps under $5 rate about 4.26 to 4.30, while apps over $10 rate lowest (4.13).
- High-rated apps get 65% positive reviews against 54% for low-rated ones.

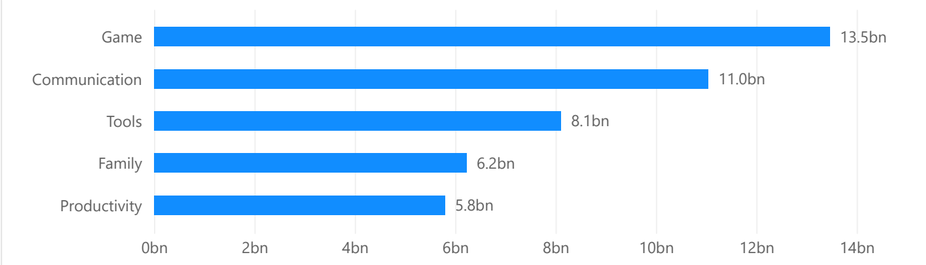
Recommendations

- Use a free model (ads or in-app purchases) for reach, and keep any paid app under $5.
- Build in high-install categories for volume, and in high-rated niches such as Events, Education and Books for quality and less competition.
- Aim for the "Everyone" content rating (82% of apps), which gives the widest audience.
- Keep app size in line with category norms (Tools about 9 MB, Games about 42 MB).
- Update regularly, since most apps were updated within the year.
- Track review sentiment, and fix recurring complaints to move from the low band to the high band.

Total Apps	free app %	apps rated 4+	Unique categories	Avg Rating
9.658K	92.2%	6.287K	33	4.17

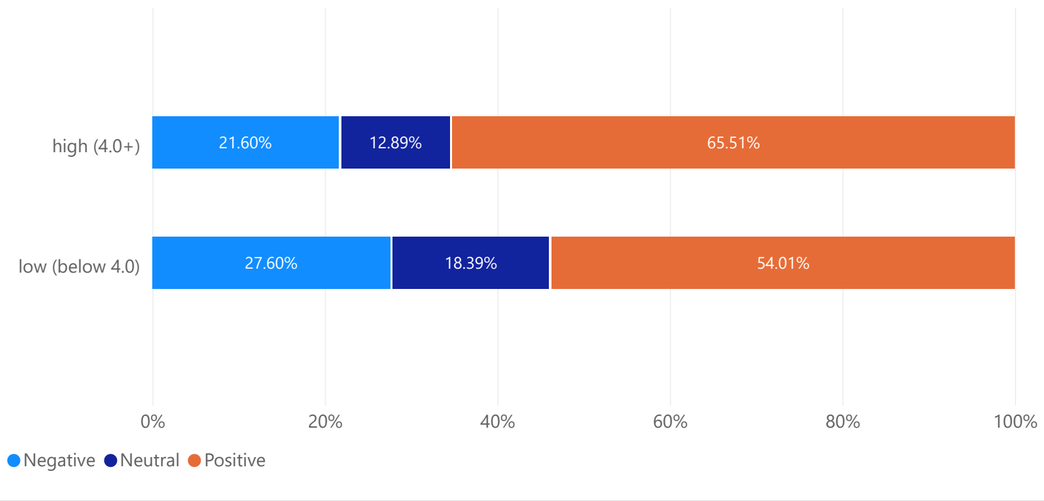
M10 Top 5 Categories By Installs

Total Installs by Category



A4 Sentiment Mix By App Rating Band

Review count and Sentiment % by review Rating Band and Sentiment



M2 Highest Rated Categories

Avg Rating by Category

